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Bark water vapor conductance varies among temperate forest tree species and is affected by flooding and stem bending

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Key words: periderm, functional traits, extreme weather events, water loss

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Abstract

Bark water vapor conductance (g_{bark}) modulates forest transpiration during droughts, when leaf transpiration is highly reduced. If disturbances such as windstorms and floods impact g_{bark}, they could affect tree performance during subsequent droughts. Bark traits, particularly lenticel traits, likely drive variation in g_{bark} and may influence the effects of disturbances on g_{bark}. We assessed variation in g_{bark} and bark traits in tree branches of 15 tree species in a temperate forest in Louisiana, USA, and performed experiments to test whether g_{bark} in tree branches was affected by bending (simulating wind) and whether g_{bark} of tree saplings was affected by soil flooding. Among tree species, mean branch g_{bark} ranged from 2.22 to 12.02 mmol m^{-2} s^{-1}. Stem bending increased g_{bark} by 23% compared to unbent controls. Although g_{bark} was unaffected by 38 days of flooding, after 69 days post-flood it was reduced by 41% compared to unflooded controls. The relationships between g_{bark} and bark traits, including bark thickness, lenticel density, and lenticel size, were inconsistent across the survey and experiments. Together, these results show that g_{bark} is variable among species and mutable. Uncovering the drivers of g_{bark} variation within and among trees, including exposure to extreme weather events, will inform projections of forest dynamics under climate change.

Introduction

Terrestrial systems will likely experience more frequent and intense extreme weather events this century associated with climate change, such as droughts, hurricanes, and floods (IPCC 2021, Zabin et al. 2022). Drought-induced forest mortality events that have occurred around the globe are expected to become more frequent (Allen et al. 2010, Senf et al. 2020). Forests are vital components of the Earth system (Lindner et al. 2010), so predicting extreme-weather-induced tree mortality is an important challenge. Although the mechanisms are not fully understood, drought-induced tree mortality is generally preceded by catastrophic loss of stem hydraulic conductance (i.e., hydraulic failure) caused by embolisms that form as stem water potential declines (Adams et al. 2017). Trees follow a sequence of water-conserving responses that slow embolism formation, such as stomatal closure and leaf shedding, but despite these responses, residual water loss through leaves and bark can determine tree water balance and survival (Ryan 2011, Wang et al. 2018, Brodribb et al. 2020, Hartmann et al. 2022).

When soil water is available, water loss through leaf stomata and cuticles accounts for around 94% of total plant water loss (Lintunen et al. 2021). However, after leaf stomata close, the proportion of water lost through the bark increases. For example, when *Pinus halepensis* trees were subjected to drought conditions, water loss from bark accounted for 64% to 78% of total water loss (Lintunen et al. 2021). Water loss through bark under severe drought conditions can lead to hydraulic failure and tree mortality (Loram-Lourenço et al. 2022). The water loss rate per unit of bark surface area, standardized by the difference in water vapor mole fraction between the air and stem section, is referred to as bark water vapor conductance (g_{bark}, mmol

m^{-2} s^{-1}). Drought performance of tropical tree saplings was highly correlated with g_{bark} (Wolfe 2020), which suggests that g_{bark} may be a useful trait for projecting drought-induced tree mortality.

In addition to impeding water loss, the outer bark of trees serves myriad functions, including facilitating gas exchange, insulating against excessive heat, serving as a barrier to pests and pathogens, and providing structural support (Lendzian 2006). Tradeoffs among these functions likely drive variation in bark traits. Previous studies have shown that g_{bark} is highly variable among tree species (Table 1). While low g_{bark} would retain water during drought, it would also limit atmospheric oxygen access and, as a result, limit stem respiration and tree growth (Langenfeld-Heyser 1997, Paine et al. 2010, Loram-Lourenço et al. 2022). Bark thickness is associated with mechanical and photosynthetic traits in stems and attached leaves (Rosell et al. 2014). Stems with higher bark thickness are assumed to have limited gas exchange and lower g_{bark}, which is supported by the few studies that have tested this (Wolfe 2020, Loram-Lourenço et al. 2022, Ávila-Lovera and Winter 2024). Since bark thickness is highly dependent on stem diameter and varies widely (Rosell 2016), and since tree species vary greatly in the stem diameter of their terminal branches (Smith et al. 2014), the ratio of bark thickness to stem radius may be a useful trait for comparing investment in g_{bark} of branches among species.

Tree bark forms lenticels, i.e., lenticular shaped structures within periderm, that function as gas-water pathways by increasing bark permeability and diffusion rates (Esau 1965, Schönherr and Ziegler 1980, Lendzian 2006, Morris and Jansen 2017). For trees in waterlogged soil, lenticels can function as sites for toxic gas efflux and O_2 influx, facilitating the entry of atmospheric O_2 and its transport to roots (Topa and McLeod 1986, Kozlowski 1986). Upon flooding, lenticel hypotrophy is formed on the stems of many tree species, which is thought to further facilitate gas fluxes (Larson et al. 1991). Among tree species in a tropical savanna, a higher density of lenticels was associated with higher g_{bark} (Loram-Lourenço et al. 2022). However, despite observations of lenticel formation, it remains untested how g_{bark} responds during and after flooding. With the formation of aerenchyma and lenticels accelerating the gas exchange, we expected that flooding would increase g_{bark} at the base of tree saplings.

Stem bending caused by wind deforms branches and alters growth dynamics (Roignant et al. 2018, Ray and Bret-Harte 2019). Considering that the cells in lenticels are loosely arranged (Rosner and Morris 2022) and that they could be disarranged through compression and extension, we predicted that bending would increase g_{bark}. Globally, the area affected and severity of intense wind storms have increased in the past decades (e.g., Suvanto et al. 2016, Peterson et al. 2019, Sulik and Kejna 2020, Nosnikov et al. 2020), which will likely continue this century (IPCC 2021, Glago 2021). The southeastern U.S., particularly, has experienced recurrent hurricanes and tropical storms that have resulted in enormous ecological disturbances and losses to the forestry sector (Chen et al. 2019, Wiener et al. 2020, Trepanier and Scheitlin 2014)). These storms are increasing in intensity and severity (Mock 2008). Assessing variation in g_{bark} and its response to disturbances like flooding and stem bending will help to project tree drought performance and ecosystem function under climate-change scenarios.

The main objectives of this study were to 1) assess variation in g_{bark} among temperate forest trees and along branches within trees; 2) evaluate the relationship between g_{bark} and bark traits (i.e., bark thickness ratio, lenticel density, and lenticel size); and 3) test whether g_{bark} is affected by stem bending and soil waterlogging. To address these objectives, we surveyed g_{bark} and bark traits among 15 tree species in an upland forest in a temperate deciduous forest, performed a bending experiment on branches of four tree species, and subjected saplings of two tree species to flood conditions in a potted-plant experiment.

Methods

g_{bark} survey

To assess how g_{bark} and bark traits varied among temperate forest trees and along branches within trees, we measured 15 common species at the Louisiana State University Agricultural Center's Burden Forest located in Baton Rouge, Louisiana (30.41, -91.11) from October 2020 to March 2021. The forest is a 60.7 ha area that is managed for research, teaching, conservation, and recreation. Mean annual precipitation and air temperature in the region are 1700 mm and 20.2°C. The warmest months are July and August, mean = 28.4°C and the coldest month is January, mean = 11.2°C (Data from NOAA Online Weather Data; US Department of Commerce). Tree species were selected based on the accessibility of individual trees and to encompass a wide phylogenetic diversity (Table 2).

For each species, we randomly selected three trees that were >10 cm DBH. One ~1m long terminal branch was cut from each tree, sealed in opaque plastic bags that were humidified with wet paper towels and brought to the laboratory for measurements. Three stem segments were cut from each branch for measurements. The segments ranged from 3 to 15 cm in length with a mean ± SD of 7.92 ± 2.38 cm (unless otherwise noted, reported values are mean ± SD). The segments were located at approximately 30, 60, and 90 cm from the apex, and were categorized as “proximal” (the closest from the bole of the tree, mean = 93.9 cm from the apex), “middle” (the second closest from the bole of the tree, mean = 58.1 cm from the apex), and “distal” (the furthest from the bole of the tree, mean = 32.1 cm from the apex), respectively, for analyses. The distance from the apex varied for each branch to avoid branch junctions and areas of damaged bark. Details of g_{bark} and trait measurements are given below.

Bending experiment

To test whether stem bending affects g_{bark}, we conducted an experiment in which isolated tree branches were bent. All samples were collected from Burden Forest. Four tree species were studied: *Magnolia grandiflora, Platanus occidentalis, Pinus elliotti, and Pinus taeda*. Three trees of each species that were >10 cm DBH were sampled. From each tree, two ~1 m length terminal branches were cut and brought to the laboratory in opaque plastic bags that were humidified with wet paper towels. One branch from each tree was subjected to a bending treatment, referred to as “bent”. The bent treatment consisted of bending the branch around a 40 cm diameter circular metal frame that was covered with a rubber tube (tread) to prevent abrasion (Fig. 1). Leaves and side branches were removed and the branch was bent 10 times around the frame, allowing it to return to its straight position between bends. To ensure that the pressure was evenly distributed across the circumference of the branch, it was rotated between each bend to alternate between areas of compression and tension. The other branch in each pair was subjected to an unbent control treatment, referred to as "control". To isolate the effect of bending from any abrasion caused by the frame, the same frame with a rubber tube was rolled along the length of each branch 10 times. To ensure that the entire circumference of the branch was evenly contacted by the frame, it was rotated between each roll. Immediately following the treatments, three segments from each branch were measured for g_{bark}, located at the distal, middle, and proximal positions as described above (mean = 32.7, 60.4, and 86.9 cm from the apex, respectively).

Flooding experiment

We used a greenhouse experiment to test how g_{bark} responds to flooding. We obtained 13-month-old sweetgum (*Liquidambar styraciflua*, moderately waterlogging tolerant) and water tupelo (*Nyssa aquatica*, extremely waterlogging tolerant) saplings from Cypress Brake tree farm (Rayville, LA, USA). At the Louisiana State University Agricultural Center's Plant Material Center in Baton Rouge, Louisiana, USA, we transplanted saplings into 2.37 L containers with a 50:50 mix of loamy topsoil sourced from Clinton Township, LA, USA and sand at least two weeks before the initiation of treatments to allow acclimation in the greenhouse. After transplanting, all saplings were maintained well-watered. Air temperature and relative humidity (RH) were measured with a temperature and RH probe (HygroVUE10, Campbell Scientific, Logan, Utah). Measurements were stored every five minutes on a datalogger (CR1000, Campbell Scientific). The air temperature during the experiment was 26.4 ± 3.1°C, and vapor pressure deficit was 0.51 ± 0.42 kPa. The heights of *L. styraciflua* and *N. aquatica* saplings were 44.1 ± 17.6 cm and 71.0 ± 10.1 cm, respectively. To quantify the effects of time since flooding on g_{bark}, we performed three rounds of a flooding treatment and an unflooded control treatment (Fig. 2). Five saplings of each species were randomly placed in one of three rounds of flooding or a control. The flood treatment was imposed by placing the pots in tubs filled with tap water 3 cm above the soil surface. The three rounds of flooding treatment were from 1 June 2022 to 8 July 2022 (37 days), 28 June 2022 to 4 August 2022 (37 days), and 8 August 2022 to 15 September 2022 (38 days). Afterward, the pots were removed from the tubs, allowed to drain freely, and maintained well-watered. This design allowed the saplings to experience various lengths of time post-flood before they were measured for g_{bark} between 15 September 2022 and 20 September 2022 (Fig. 2). One *L. styraciflua* sapling was lost in the Round 2 flooding treatment. Two segments that were 6.53 ± 1.67 cm in length (range = 2–10 cm) were cut from the stems near the distal and proximal end of each sapling to measure g_{bark} (mean = 19.5 and 48.8 cm from the apex, respectively). The total number of segments measured in the flooding experiment was 78 from 39 saplings.

Measurement of g_{bark}

We used the 'mass loss' method to measure the g_{bark} of sampled stem segments (Wolfe 2021). We sealed the cut ends of the segments and leaf scars with 5 or more layers of parafilm to stop evaporation and placed the segments on a wire-mesh frame under a fan. At approximately one-hour intervals, we weighed the stem segments on an analytical balance (model CP225D, Sartorius, Goettingen, Germany) for seven hours.

To calculate g_{bark}, the rate of water loss (g s^{-1}) was assessed as the slope of an ordinary least squares regression of segment mass over time. The water loss rate was standardized by the exposed bark surface area (g m^{-2} s^{-1}), which was estimated assuming a cylindrical shape from diameter and length measurements. The water loss rate was converted to units typically used for transpiration (E; mmol m^{-2} s^{-1}) (1 g water = 55.51 mmol), which was divided by the driving force of water vapor, the mole fraction difference (D; dimensionless) to obtain the water vapor conductance (g_{bark} = E/D; mmol m^{-2} s^{-1}). The stem segments were assumed to be at water vapor saturation and in equilibrium with air temperature (see Wolfe 2020). We measured D with a CS500 sensor (Campbell Scientific, Logan, UT, USA). Measurements of air temperature and relative humidity were taken every 30 seconds and averaged at 5-minute intervals, and recorded on a CR1000 datalogger (Campbell Scientific) to compute D. During the g_{bark} measurement, the air temperature was 20.4 ± 0.3℃ and D was 0.009 ± 0.0008.

Measurement of bark traits

After measuring g_{bark}, bark traits were measured on each stem segment. Stem diameter and bark thickness were measured with calipers on both the distal and basal end of each segment. Bark included all tissue radially outward of the vascular cambium. The bark thickness ratio was calculated as two times the bark thickness divided by stem diameter (Hoffmann et al. 2012).

For the g_{bark} survey samples, high-resolution images of the stem segments were taken using a Canon CanoScan LiDE 400 flatbed scanner. Within a rectangular area on the surface of each sample, the total number of lenticels was counted, and the area of each lenticel (a measurement of lenticel size) was measured using ImageJ (Schneider et al. 2012). For each segment, lenticel density was calculated as the number of lenticels divided by the area of the stem surface measured and lenticel size was taken as the average area of the lenticels.

For the flooding experiment, lenticels were tallied on the exposed area of the stem segment not covered by the parafilm, and lenticel density was calculated by dividing the total number of lenticels by the exposed surface area. Lenticel width (a proxy for lenticel size, perpendicular to the longest dimension of a lenticel) was measured on 10 randomly selected lenticels on each segment with calipers under a 10× lens and averaged for each segment. All lenticel measurements were performed on primary lenticels, whereas secondary lenticels, which are often obscured by successive periderm layers (Rosner and Morris 2022), were excluded due to the difficulty of observing them.

Data analysis

All analyses were performed in R version 4.2.2 (R Core Team 2022). We transformed the response variable when needed to meet the underlying model assumptions. We set the significance level (alpha) to 0.05.

For the g_{bark} survey, a two-way ANOVA was performed to test the effects of species and segment position on g_{bark}. The response variable was square root transformed g_{bark} and the explanatory variables were species, segment position, and their interaction. Two planned contrasts were used to compare the means of g_{bark}; one compared among species and one compared among segment positions. We tested for relationships between species-level bark traits and g_{bark} using ordinary least squares regression. We also used ordinary least squares regression to test for relationships within species between g_{bark} and traits. For eight species with enough data on lenticel density and lenticel size (i.e., at least five out of the nine stem segments measured had at least one lenticel), the response variable was g_{bark}, and the explanatory variables were lenticel size, lenticel density, the interaction of lenticel size and lenticel density, and bark thickness ratio. For the other seven species without enough data on lenticel traits, the response variable was g_{bark}, and the explanatory variable was bark thickness ratio only.

For the bending experiment, we performed a three-way ANOVA with an error term to account for the split-plot design, where species served as the whole plot factor, and bending treatments were the split-plot factor, which was nested in by the segment positions. The response variable was the natural log transformed g_{bark}, with bending treatment, species, segment position, and their interaction, included as the fixed effects. Planned contrasts were used to compare g_{bark} between the two levels of bending treatment, among species, and across segment positions.

For the flooding experiment, we performed a three-way ANOVA to test the effects of flooding treatment, species, and segment positions on g_{bark}. The response variable was natural log transformed g_{bark}, and the explanatory variables were flooding treatment, species, segment position, and their interaction. We used planned contrasts to compare g_{bark} among post-flood treatments, species, and segment positions. The relationships between g_{bark} and lenticel density, lenticel size, and bark thickness ratio for each species were tested by ordinary least squares regression. The response variable was natural log transformed g_{bark} for *L. styraciflua* and *N.*

aquatica separately, and the explanatory variables were lenticel density, lenticel size, and bark thickness ratio. Additionally, for each species, we performed ANOVA to test the effect of flooding treatment on lenticel density and lenticel size. The response variables were lenticel density or lenticel size of *L. styraciflua* and *N. aquatica* separately, and the explanatory variable was flooding treatments. If the interactions are not significant for the tests, we reported the p-values of the final models without interaction.

Results

g_{bark} survey

Among 135 stem segments measured in the g_{bark} survey, the mean g_{bark} was 3.84 ± 1.43 mmol m^{-2} s^{-1}, with a range of 0.87 to 9.20 mmol m^{-2} s^{-1}. g_{bark} varied among species and segment positions (Fig. 3; $F_{14,90} = 12.80$, $p < 0.001$; and $F_{2,90} = 5.54$, $p < 0.01$; respectively), and not their interaction ($F_{28,90} = 0.61$, $p = 0.93$). Among species, g_{bark} ranged from 2.22 ± 0.93 mmol m^{-2} s^{-1} in *Fagus grandifolia* to 6.31 ± 1.69 mmol m^{-2} s^{-1} in *Acer negundo*. Segment positions ranged in g_{bark} from 3.48 ± 1.23 mmol m^{-2} s^{-1} in the distal position (closest from the apex) to 4.11 ± 1.53 mmol m^{-2} s^{-1} in the proximal position (furthest from the apex) (Fig. 4). Among species, g_{bark} did not scale with lenticel density, lenticel size, or bark thickness ratio (Fig. S1).

The intraspecific relationships between g_{bark} and bark traits differed among species (Table S1). Only two variables were significantly related to g_{bark}; bark thickness ratio for *Acer negundo* (P = 0.04) and lenticel density for *Betula nigra* (P = 0.02). Both were negatively related to g_{bark} For seven out of the eight species on which we encountered lenticels, lenticel density trended negatively with g_{bark}. No consistent trends were found for lenticel size or bark thickness ratio.

Bending experiment

In the bending experiment, across all species and segment positions, g_{bark} averaged 5.55 ± 2.19 mmol m^{-2} s^{-1}, with a range of 2.66 to 13.62 mmol m^{-2} s^{-1}. The ANOVA showed that g_{bark} was significantly affected by bending treatment and species, but not segment position or any interactions (Fig. 5; $F_{1,59} = 4.44$, $p = 0.03$; $F_{3,59} = 10.57$, $p < 0.001$; $F_{2,59} = 0.67$, $p = 0.52$; and $F_{6,59} = 0.16$, $p = 0.99$; respectively). On average, there was an increase of 1.13 mmol m^{-2} s^{-1} g_{bark} from the control segments to the bent segments (Fig. 5A). g_{bark} was similar across species except for *P. occidentalis*, where it was significantly lower (Fig. 5B).

Flooding experiment

Among all segments of both species in the flooding experiment, g_{bark} averaged 8.10 ± 7.44 mmol m^{-2} s^{-1} with a range of 1.54 to 51.21 mmol m^{-2} s^{-1} (Fig. 6). g_{bark} differed between species (Fig. 6B; $F_{1,62} = 116.34$, $p < 0.001$), with values of 4.37 ± 2.96 mmol m^{-2} s^{-1} in *N. aquatica* and 12.02 ± 8.68 mmol m^{-2} s^{-1} in *L. styraciflua*. g_{bark} varied among post-flood durations (Fig. 6A; $F_{3,62} = 4.43$, $p < 0.01$). It ranged from 6.32 ± 4.45 mmol m^{-2} s^{-1} in the 69-day post-flood group to 10.70 ± 11.30 mmol m^{-2} s^{-1} in the control group. g_{bark} also varied between segment positions (Fig. 6C; $F_{1,62} = 62.05$, $p < 0.001$), with proximal stem segments, on average, over twice as high as distal stem segments (11.10 ± 9.09 versus 5.10 ± 3.34 mmol m^{-2} s^{-1}). No interactions were significant.

The relationships between g_{bark} and bark traits differed between species (Table 3). In *L. styraciflua*, g_{bark} decreased as lenticel density increased ($t_{36} = -3.92$, $p < 0.001$). In *N. aquatica*, g_{bark} decreased as the bark thickness ratio increased ($t_{38} = -4.76$, $p < 0.001$). Whereas g_{bark} did not scale with other traits, the decreasing trend in g_{bark} when the bark thickness ratio increased was consistent for both species (Table 3).

During the flooding experiment, the lenticel size of *L. styraciflua* significantly differed among flooding treatments ($F_{3,34} = 5.83$, $p < 0.01$). The lenticel size of *L. styraciflua* was 0.44 ± 0.12 mm in the control group, 0.57 ± 0.12 mm in the 0 day post-flood duration group, 0.64 ± 0.15 mm in the 42 days post-flood duration group, and 0.68 ± 0.20 mm in the 69 days post-flood duration group. The lenticel density of *L. styraciflua* and lenticel size and lenticel density of *N. aquatica* did not differ among flooding treatments.

Discussion

g_{bark} variation among species

Tree species differed in g_{bark} within our survey in a temperate deciduous forest in the southeastern USA and within both of our experiments. In our survey, the mean and range of g_{bark} (Fig. 1) were similar to those of previous studies in various biomes (Table 1). Consistently, g_{bark} variation is high among species within biomes and low among biomes, indicating that biome is not informative in predicting g_{bark}. Nonetheless, g_{bark} contains phylogenetic signals (Ávila-Lovera and Winter 2024) and is an informative species-level trait (Wolfe 2020, Loram-Lourenço et al. 2022). However, like other functional traits, g_{bark} varies within individuals and in response to environmental conditions. Among co-occurring species within biomes, the variation in g_{bark} along with other traits indicates different ecological strategies (niches) to avoid interspecies competition and maintain diversity (Chase and Leibold 2003, Kraft et al. 2008).

g_{bark} variation along stems

Our survey and flooding experiment both showed that stem segments further from the apex had higher g_{bark} (Figs. 4, 6C). In contrast, the bending experiment found no differences in g_{bark} among segment positions (Fig 5C). However, there was a trend for lower g_{bark} in distal segments in the bending experiment, so the lack of effect is probably due to a smaller sample size. In contrast to our results, Wolfe (2020) found a weak and marginally significant trend for g_{bark} to decline from the distal to proximal positions of tropical tree saplings, from 3.85 mmol m^{-2} s^{-1} at 4 cm from the apex to 2.94 mmol m^{-2} s^{-1} at 250 cm from the apex. Whereas we measured terminal branches of large trees in the survey and bending experiment and small saplings in the flooding experiment, Wolfe (2020) measured across the entire stem length of large saplings. The factors that underlie these divergent results are unclear; however, they may be related to ontogeny and sampling design.

Associations between bark traits and g_{bark}

We found no clear relationship between g_{bark} and lenticel density. Among tree species in our survey, g_{bark} was unrelated to lenticel density (Fig. S1), whereas within most species there was an insignificant trend for a negative relationship between g_{bark} and lenticel density (Table S1).

In the flooding experiment, the relationships between g_{bark} and bark traits were not consistent, with only lenticel density of *L. styraciflua* and bark thickness ratio of *N. aquatica* being significantly related to g_{bark}. Lenticel size was inversely related or showed a negative trend with lenticel density in most species (Table S2). Lenticel size did not affect g_{bark} in any of the species. However, Rosner and Morris (2022) suggested that lenticel size was positively related to g_{bark} in *Picea abies*. Overall, our results contrast those of Loram-Lourenço et al. (2022), who found that g_{bark} was negatively related to relative outer bark thickness and positively related to lenticel density among terminal branches of 10 tree species in a Brazilian savanna.

Even though substantial evidence indicates lenticels are more permeable to water vapor and oxygen than phellem areas without lenticels (Schönherr and Ziegler 1980, Groh et al. 2002) and lenticels facilitate tree survival during long-term flooding (Larson et al. 1991), plants regulate lenticel properties seasonally (Lendzian 2006), which implies the mechanisms behind lenticels controlling bark gas exchange are complex and not sufficiently explored. The considerable variability of lenticel structure and abundance and the probability of waxes covering the inner lenticel surface may weaken the associations between lenticel traits and g_{bark} (Rosner and Morris 2022). We measured g_{bark} outside of the growing season, when lenticels may have been partly occluded by an intact, densely packed, and suberized cell layer (Rosner and Morris 2022), which would diminish the relationship between g_{bark} and lenticel density and size. A similar structure was found on potato tuber lenticels to reduce the conductance to water vapor (Hayward 1974). However, the seasonality of g_{bark} and its relationship with lenticel structure is not clear and requires further research. Moreover, as rhytidome forms, cell division within primary lenticels often slows, causing the lenticels to become less distinct (Crang et al. 2018), which would further reduce their relationship with g_{bark}. Future studies that control for bark structure rather than distance from the apex would have more ability to detect relationships between g_{bark} and lenticel traits.

In the flooding experiment, g_{bark} decreased with higher bark thickness ratio (Table 3). This result is similar to those of Wolfe (2020), Loram-Lourenço et al. (2022), and Ávila-Lovera and Winter (2024). Commonly, bark permeability is thought to decrease with bark thickness (Paine et al. 2010). Tree species that grow in drier and hotter environments tend to have thicker bark (Rosell 2016), which may reflect adaptation for reducing water loss. However, our g_{bark} survey did not find consistent relationships between the bark thickness ratio and $g_{bark,}$ within or among species (Fig. S1, Table S1), which contrasted with the previous studies (Wolfe 2020, Loram-Lourenço et al. 2022, Ávila-Lovera and Winter 2024), probably because the average bark thickness ratio of trees from the g_{bark} survey was consistently low, 0.09 ± 0.03 with a variance of 0.0008. The limited variability of bark thickness ratio of the g_{bark} survey made it difficult to detect significant relationships. Also, Loram-Lourenço et al. (2022) measured the outer bark thickness, which is the barrier to water vapor and is expected to relate more closely to g_{bark} compared to the total bark thickness in the current survey. Ávila-Lovera and Winter (2024) found that g_{bark} decreased with bark thickness ratio among tropical tree species, however, they measured g_{bark} in stem segments located 10 cm from the apex. This placement presumably included areas of primary growth, as indicated by the presence of stomata described by the authors, as opposed to the secondary growth segments we collected, which may have led to the divergent results.

Effects of stem bending on g_{bark}

Directly after stems were bent, they had significantly higher g_{bark} than unbent controls (Fig. 5A). Although no macroscopic bark damage was observed, it is possible that bending caused undetected damage to the outer bark that was obscure or microscopic. For example, compression and expansion of suberin and wax layers in periderm cells (including those in the lenticels) or abrasion between successive layers in rhytidome could have increased g_{bark}. Similar to our results, when leaves were subjected to wind, cuticle damage caused higher minimum vapor conductance in conifer trees (van Gardingen et al. 1991). It is unclear to what extent our bending experiment simulated stem bending caused by wind; however, considering that none of the stems broke during the experiment and that stem breakage is a common response to cyclones (Gleason et al. 2008), our experiment likely captured realistic bending stress that occurs during windstorms. Wind damage can affect forest vulnerability to subsequent droughts in multiple ways; for example, when hurricanes create canopy gaps, environmental conditions favor fast-growing tree species that are vulnerable to drought (Smith-Martin et al. 2022). Our results suggest that increased g_{bark} is an additional mechanism by which windstorms can make forests vulnerable to droughts. All else equal, residual water loss from trees will increase in proportion to g_{bark} and the drought timespan to reach critical thresholds of stem water potential will decrease. Distinguishing the wind conditions that affect g_{bark} and how they vary among trees will require further research.

Effects of flooding on g_{bark} *and lenticels*

Contrary to our expectations, we found that 38 days of flooding did not cause increased g_{bark} (Fig 6A). Instead, g_{bark} decreased during the recovery process (Fig. 6A). Based on extensive observations of flooding causing saplings to form hypertrophied lenticels near the waterline on their stems (Esau 1965, Schaffer 1998, Shimamura et al. 2010), and evidence that hypertrophied lenticels are highly permeable (Rennenberg et al. 1997), we expected proximal stem segments to have particularly high g_{bark} in response to flooding. However, the lack of this response may reflect that the saplings were well adapted to the low soil oxygen availability in our flood treatment, which would preclude a stress response such as increased g_{bark}. *L. styraciflua* and *N. aquatica* are described as moderately flood-tolerant (survives waterlogging or saturated soils for 30 consecutive days during the growing season) and extremely flood-tolerant (survives deep, prolonged waterlogging for more than one year), respectively (Niinemets and Valladares 2006). Constitutively high g_{bark} near the stem base (i.e., proximal position) may be an adaptation for flood tolerance. Since hypertrophied lenticels develop over several days or weeks, constitutively high g_{bark} would help to reduce root oxidative stress at the onset of soil waterlogging. This is supported by the result that g_{bark} was higher in proximal segments than distal segments across all treatments (Fig. 6C). This result contrasts with those of Wolfe (2020), who found that g_{bark} varied only marginally from distal to the proximal positions of saplings in seasonally dry tropical forests that are not prone to flooding. The proximal position of these two flood-tolerant species also had relatively high g_{bark} compared to the saplings measured by Wolfe (2020), which is the only other report of g_{bark} in saplings that we are aware of. However, constitutively high g_{bark} would be disadvantageous during drought conditions, which may contribute to the tradeoff between flood tolerance and drought tolerance among tree species (Niinemets and Valladares 2006).

The effect of flooding on lenticels varied between the two tree species. In *L. styraciflua*, lenticel size increased after the 38-day flooding period and continuously increased with a longer post-flood duration, with no corresponding increase in g_{bark}. This is consistent with results from Angeles et al. (1986), who observed the formation of hypertrophied lenticels in *Ulmus americana* seedlings as soon as five days after flooding. The morphological changes included more active phellogen and elongated cork, phellem, and cells, enhancing the gas exchange rate of the bark (Angeles et al. 1986). No changes were found in the lenticel size in *N. aquatica*, which likely possesses adaptations that enable it to grow in flooded environments without morphological changes like lenticel hypertrophy.

After a period of 69 days post-flood, g_{bark} declined below that of the control plants (Fig. 6A), contrasting with our expectations. The processes that drove this g_{bark} decline are unclear. One possibility is that the flood treatment induced early dormancy in the saplings. In many tree species, flooding causes early leaf senescence and physiological dormancy (Pereira and Kozlowski 1977, Vu and Yelenosky 1991). Winter dormancy is associated with the formation of a lenticel closing layer composed of densely packed suberized cells (Langenfeld-Heyser 1997), which would reduce g_{bark}. Therefore, the saplings that were exposed to flooding in the early summer months could have entered a dormant state by the time they were measured in the early fall, while the control saplings were still active with unoccluded lenticels leading to the observed higher g_{bark}. Since lower g_{bark} is associated with higher drought performance

(Wolfe 2020, Loram-Lourenço et al. 2022), the reduced g_{bark} in the post-flood treatment would be expected to enhance drought performance compared to the control trees that were not exposed to flood conditions. However, it is unclear how long the trees would maintain a post-flood g_{bark} reduction or whether other effects of flooding would diminish drought performance.

Conclusions

We found that g_{bark} varied widely among tree species in a temperate forest, consistent with studies in a wide range of biomes. The lack of consistent relationships between g_{bark} and bark traits such as lenticel density, lenticel size, and bark thickness point to more complex drivers of g_{bark} variation such as the structure and composition of periderms and lenticels and their dynamics. Stem bending and flooding both affected g_{bark}. This suggests that g_{bark} dynamics can mediate the interactive effects of extreme weather events on trees and forests. Extreme weather events are likely to become more common this century (Beniston et al. 2007, Konisky et al. 2015), so resolving the factors that lead to g_{bark} variation, its dynamics, and its impacts on tree performance should be a research priority.

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Authors' contributions

B.T.W., G.R., and Y.Z designed and performed the research; Y.Z., B.T.W., and V.D. analyzed the data. Y.Z. wrote the paper and B.T.W. revised it. All authors reviewed and approved the final manuscript.

Data availability statement

Data are available from the corresponding author upon request.

Conflict of interest

The authors declare that they have no conflict of interest.

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Figure legends

Figure 1 Schematic figure for bending experiment. Two ~1 m length terminal branches were collected from three trees of four species. One branch from each tree was subjected to an unbent control treatment (A), referred to as "control". The frame was rolled along the branch to account for potential abrasion. The other branch was subjected to a bending treatment (B), referred to as "bent". We bent the branch 10 times around the frame and allowed it to return to its straight position. The treatments were applied for 10 times. The branch was rotated between each treatment to ensure evenly distributed pressure.

Figure 2 Schematic figure for flooding experiment. Panel (A) indicates the control treatment and panel (B) indicates the flooding treatments with consistent flood durations and variable post-flood durations. Five saplings of each species were placed in each round of flooding and the control. In total 40 saplings were applied with one *L. styraciflua* sapling was lost in the Round 2 flooding treatment; thus 39 saplings were harvested. The saplings experienced different lengths of time post-flood and were measured for g_{bark}.

Figure 3 Boxplot of bark vapor conductance (g_{bark}) among tree species surveyed in a temperate forest. Each box extends to 25% and 75% quartiles of values and is bisected by the median. Whiskers extend to the most extreme values within 1.5 times the interquartile range. Circles outside of the whiskers represent outliers. Differing letters above each boxplot represent statistically different square root transformed g_{bark} values between combinations, based on planned contrasts.

Figure 4 Boxplot of bark vapor conductance (g_{bark}) among stem segment positions surveyed in a temperate forest. Boxes, whiskers, and differing letters are drawn as in Fig. 3. Blue lines represent pairs that had higher g_{bark} in the proximal segments than in the distal segments, and red lines, vice versa. Differing letters above each boxplot represent statistically different square root transformed g_{bark} values between combinations, based on planned contrasts.

Figure 5 Boxplots of bark vapor conductance (g_{bark}) in the bending experiment grouped by (A) treatment, (B) species, and (C) segment position. Boxes and whiskers are drawn as in Fig. 3. In panel (A) lines connect g_{bark} values of segment pairs (i.e., same tree and segment position). Blue lines represent pairs that had higher g_{bark} in the bent treatment than in the control treatment and red lines, vice versa. Differing letters above each boxplot represent statistically different natural log transformed g_{bark} values between combinations, based on planned contrasts. The letters "NS" in panel (C) indicate natural log transformed g_{bark} did not differ between segment positions in this experiment.

Figure 6 Boxplots of bark vapor conductance (g_{bark}) in the flood experiment grouped by (A) treatment, (B) species, and (C) segment position. Boxes and whiskers are drawn as in Fig. 3. Differing letters above each boxplot represent statistically different natural log transformed g_{bark} values between combinations, based on planned contrasts.

Bark water vapor conductance varies among temperate forest tree species and is affected by flooding and stem bending

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Key words: periderm~~bark~~, functional traits, extreme weather events, water loss

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Abstract

Bark water vapor conductance (g_{bark}) modulates forest transpiration during droughts, when leaf transpiration is highly reduced. If disturbances such as windstorms and floods impact g_{bark}, they could affect tree performance during subsequent droughts. Bark traits, particularly lenticel traits, likely drive variation in g_{bark} and may influence the effects of disturbances on g_{bark}. We assessed variation in g_{bark} and bark traits in tree branches of 15 tree species in a temperate forest in Louisiana, USA, and performed experiments to test whether g_{bark} in tree branches was affected by bending (simulating wind) and whether g_{bark} of tree saplings was affected by soil flooding. Among tree species, mean branch g_{bark} ranged from 2.22 to 12.02 mmol m^{-2} s^{-1}. Stem bending increased g_{bark} by 23% compared to unbent controls. Although g_{bark} was unaffected by 38 days of flooding, after 69 days post-flood it was reduced by 41% compared to unflooded controls. The relationships between g_{bark} and bark traits, including bark thickness, lenticel density, and lenticel size, were inconsistent across the survey and experiments. Together, these results show that g_{bark} is variable among species and mutable. Uncovering the drivers of g_{bark} variation within and among trees, including exposure to extreme weather events, will inform projections of forest dynamics under climate change.

Introduction

Terrestrial systems will likely experience more frequent and intense extreme weather events this century associated with climate change, such as droughts, hurricanes, and floods (IPCC 2021, Zabin et al. 2022). Drought-induced forest mortality events that have occurred around the globe are expected to become more frequent (Allen et al. 2010, Senf et al. 2020). Forests are vital components of the Earth system (Lindner et al. 2010), so predicting extreme-weather-induced tree mortality is an important challenge. Although the mechanisms are not fully understood, drought-induced tree mortality is generally preceded by catastrophic loss of stem hydraulic conductance (i.e., hydraulic failure) caused by embolisms that form as stem water potential declines (Adams et al. 2017). Trees follow a sequence of water-conserving responses that slow embolism formation, such as stomatal closure and leaf shedding, but despite these responses, residual water loss through leaves and bark can determine tree water balance and survival (Ryan 2011, Wang et al. 2018, Brodribb et al. 2020, Hartmann et al. 2022).

When soil water is available, water loss through leaf stomata and cuticles accounts for around 94% of total plant water loss (Lintunen et al. 2021). However, after leaf stomata close, the proportion of water lost through the bark increases. For example, when *Pinus halepensis* trees were subjected to drought conditions, water loss from bark accounted for 64% to 78% of total water loss (Lintunen et al. 2021). Water loss through bark under severe drought conditions can lead to hydraulic failure and tree mortality (Loram-Lourenço et al. 2022). The water loss rate per unit of bark surface area, standardized by the difference in water vapor mole fraction between the air and stem section, is referred to as bark water vapor conductance (g_{bark}, mmol

m^{-2} s^{-1}). Drought performance of tropical tree saplings was highly correlated with g_{bark} (Wolfe 2020), which suggests that g_{bark} may be a useful trait for projecting drought-induced tree mortality.

In addition to impeding water loss, the outer bark of trees serves myriad functions, including facilitating gas exchange, insulating against excessive heat, serving as a barrier to pests and pathogens, and providing structural support (Lendzian 2006). Tradeoffs among these functions likely drive variation in bark traits. Previous studies have shown that g_{bark} is highly variable among tree species (~~Table 1~~Table 1). While low g_{bark} would retain water during drought, it would also limit atmospheric oxygen access and, as a result, limit stem respiration and tree growth (Langenfeld-Heyser 1997, Paine et al. 2010, Loram-Lourenço et al. 2022). Bark thickness is associated with mechanical and photosynthetic traits in stems and attached leaves (Rosell et al. 2014). Stems with higher bark thickness are assumed to have limited gas exchange and lower g_{bark}, which is supported by the few studies that have tested this (Wolfe 2020, Loram-Lourenço et al. 2022, Ávila-Lovera and Winter 2024). Since bark thickness is highly dependent on stem diameter and varies widely (Rosell 2016),~~Since bark thickness is highly dependent on stem diameter,~~ and since tree species vary greatly in the stem diameter of their terminal branches (Smith et al. 2014), the ratio of bark thickness to stem radius may be a useful trait for comparing investment in g_{bark} of branches among species.

Tree bark forms lenticels, i.e., lenticular shaped structures within~~on the outer~~ periderm, that function as gas-water pathways by increasing bark permeability and diffusion rates (Esau 1965, Schönherr and Ziegler 1980, Lendzian 2006, Morris and Jansen 2017). For trees in waterlogged soil, lenticels can function as sites for toxic gas efflux and O_2 influx, facilitating the entry of atmospheric O_2 and its transport to roots (Topa and McLeod 1986, Kozlowski 1986). Upon flooding, lenticel hypotrophy is formed on the stems of many tree species, which is thought to further facilitate gas fluxes (Larson et al. 1991). Among tree species in a tropical savanna, a higher density of lenticels was associated with higher g_{bark} (Loram-Lourenço et al. 2022). However, despite observations of lenticel formation, it remains untested how g_{bark} responds during and after flooding. With the formation of aerenchyma and lenticels accelerating the gas exchange, we expected that flooding would increase g_{bark} at the base of tree saplings.

Stem bending caused by wind deforms branches and alters growth dynamics (Roignant et al. 2018, Ray and Bret-Harte 2019). Considering that the cells in lenticels are loosely arranged (Rosner and Morris 2022) and that they could be disarranged through compression and extension, we predicted that bending would increase g_{bark}. Globally, the area affected and severity of intense wind storms have increased in the past decades (e.g., Suvanto et al. 2016, Peterson et al. 2019, Sulik and Kejna 2020, Nosnikov et al. 2020), which will likely continue this century (IPCC 2021, Glago 2021). The southeastern U.S., particularly, has experienced recurrent hurricanes and tropical storms that have resulted in enormous ecological disturbances and losses to the forestry sector (Chen et al. 2019, Wiener et al. 2020, Trepanier and Scheitlin 2014)). These storms are increasing in intensity and severity (Mock 2008). Assessing variation in g_{bark} and its response to disturbances like flooding and stem bending will help to project tree drought performance and ecosystem function under climate-change scenarios.

The main objectives of this study were to 1) assess variation in g_{bark} among temperate forest trees and along branches within trees; 2) evaluate the relationship between g_{bark} and bark traits (i.e., bark thickness ratio, lenticel density, and lenticel size); and 3) test whether g_{bark} is affected by stem bending and soil waterlogging. To address these objectives, we surveyed g_{bark} and bark traits among 15 tree species in an upland forest in a temperate deciduous forest, performed a bending experiment on branches of four tree species, and subjected saplings of two tree species to flood conditions in a potted-plant experiment.

Methods

g_{bark} survey

To assess how g_{bark} and bark traits varied among temperate forest trees and along branches within trees, we measured 15 common species at the Louisiana State University Agricultural Center's Burden Forest located in Baton Rouge, Louisiana (30.41, -91.11) from October 2020 to March 2021. The forest is a 60.7 ha area that is managed for research, teaching, conservation, and recreation. Mean annual precipitation and air temperature in the region are 1700 mm and 20.2°C. The warmest months are July and August, mean = 28.4°C and the coldest month is January, mean = 11.2°C (Data from NOAA Online Weather Data; US Department of Commerce). Tree species were selected based on the accessibility of individual trees and to encompass a wide phylogenetic diversity (~~Table 2~~Table 2).

For each species, we randomly selected three trees that were >10 cm DBH. One ~1m long terminal branch was cut from each tree, sealed in opaque plastic bags that were humidified with wet paper towels and brought to the laboratory for measurements. Three stem segments were cut from each branch for measurements. The segments ranged from 3 to 15 cm in length with a~~were~~ mean ± SD of~~=~~ 7.92 ± 2.38 cm (unless otherwise noted, reported values are mean ± SD)~~.) and range = 3–15 cm in length.~~ The segments were located at approximately 30, 60, and 90 cm from the apex, and were categorized as "proximal" (the closest from the bole of the tree, mean = 93.9 cm from the apex), "middle" (the second closest from the bole of the tree, mean = 58.1 cm from the apex), and "distal" (the furthest from the bole of the tree, mean = 32.1 cm from the apex), respectively, for analyses. The distance from the apex varied for each branch to avoid branch junctions and areas of damaged bark. Details of g_{bark} and trait measurements are given below.

Bending experiment

To test whether stem bending affects g_{bark}, we conducted an experiment in which isolated tree branches were bent. All samples were collected from Burden Forest. Four tree species were studied: *Magnolia grandiflora, Platanus occidentalis, Pinus elliotti, and Pinus taeda*. Three trees of each species that were >10 cm DBH were sampled. From each tree, two ~1 m length terminal branches were cut and brought to the laboratory in opaque plastic bags that were humidified with wet paper towels. One branch from each tree was subjected to a bending treatment, referred to as "bent". The bent treatment consisted of bending the branch around a 40 cm diameter circular metal frame that was covered with a rubber tube (tread) to prevent abrasion (Fig. 1). Leaves and side branches were removed and the~~The~~ branch was bent 10 times around the frame, allowing it~~ and allowed~~ to return to its straight position between bends~~.~~ To ensure that the pressure was evenly distributed across the circumference of the branch, it was rotated between each bend to alternate between areas of compression and tension. The other branch in each pair was subjected to an unbent control treatment, referred to as "control". To isolate the effect of bending from any abrasion caused by the frame, the same frame with a rubber tube was rolled along the length of each branch 10 times. To ensure that the entire circumference of the branch was evenly contacted by the frame, it was rotated between each roll. Immediately following the treatments, three segments from each branch were measured for g_{bark}, located at the distal, middle, and proximal positions as described above (mean = 32.7, 60.4, and 86.9 cm from the apex, respectively).

Flooding experiment

We used a greenhouse experiment to test how g_{bark} responds to flooding. We obtained 13-month-old sweetgum (*Liquidambar styraciflua*, moderately waterlogging tolerant) and water tupelo (*Nyssa aquatica*, extremely waterlogging tolerant) saplings from Cypress Brake tree farm (Rayville, LA, USA). At the Louisiana State University Agricultural Center's Plant Material Center in Baton Rouge, Louisiana, USA, we transplanted saplings into 2.37 L containers with a 50:50 mix of loamy topsoil sourced from Clinton Township, LA, USA and sand at least two weeks before the initiation of treatments to allow acclimation in the greenhouse. After transplanting, all saplings were maintained well-watered. Air temperature and relative humidity (RH) were measured with a temperature and RH probe (HygroVUE10, Campbell Scientific, Logan, Utah). Measurements were stored every five minutes on a datalogger (CR1000, Campbell Scientific). The air temperature during the experiment was 26.4

± 3.1°C, and vapor pressure deficit was 0.51 ± 0.42 kPa. The heights of *L. styraciflua* and *N. aquatica* saplings were 44.1 ± 17.6 cm and 71.0 ± 10.1 cm, respectively. To quantify the effects of time since flooding on g_{bark}, we performed three rounds of a flooding treatment and an unflooded control treatment (Fig. 2). Five saplings of each species were randomly placed in one of three rounds~~each round~~ of flooding or a~~and the~~ control. The flood treatment was imposed by placing the pots in tubs filled with tap water 3 cm above the soil surface. The three rounds of flooding treatment were from 1 June 2022 to 8 July 2022 (37 days), 28 June 2022 to 4 August 2022 (37 days), and 8 August 2022 to 15 September 2022 (38 days). Afterward, the pots were removed from the tubs, allowed to drain freely, and maintained well-watered. This design allowed the saplings to experience various lengths of time post-flood before they were measured for g_{bark} between 15 September 2022 and 20 September 2022 (Fig. 2). One *L. styraciflua* sapling was lost in the Round 2 flooding treatment. Two segments that were 6.53 ± 1.67 cm in length (range = 2–10 cm) were cut from the stems near the distal and proximal end of each sapling to measure g_{bark} (mean = 19.5 and 48.8 cm from the apex, respectively). The total number of segments measured in the flooding experiment was 78 from 39 saplings.

Measurement of g_{bark}

We used the 'mass loss' method to measure the g_{bark} of sampled stem segments (Wolfe 2021). We sealed the cut ends of the segments and leaf scars with 5 or more layers of parafilm to stop evaporation and placed the segments on a wire-mesh frame under a fan. At approximately one-hour intervals, we weighed the stem segments on an analytical balance (model CP225D, Sartorius, Goettingen, Germany) for seven hours.

To calculate g_{bark}, the rate of water loss (g s^{-1}) was assessed as the slope of an~~with~~ ordinary least squares regression of segment mass over time.~~.~~ The water loss rate was standardized by the exposed bark surface area (g m^{-2} s^{-1}), which was estimated assuming a cylindrical shape from diameter and length measurements. The water loss rate was converted to units typically used for transpiration (E; mmol m^{-2} s^{-1}) (1 g water = 55.51 mmol), which was divided by the driving force of water vapor, the mole fraction difference (D; dimensionless) to obtain the water vapor conductance (g_{bark} = E/D; mmol m^{-2} s^{-1}). The stem segments were assumed to be at water vapor saturation and in equilibrium with air temperature (see Wolfe 2020). We measured D with a CS500 sensor (Campbell Scientific, Logan, UT, USA). Measurements of air temperature and relative humidity were taken every 30 seconds and averaged at 5-minute intervals, and recorded on a CR1000 datalogger (Campbell Scientific) to compute D. During the g_{bark} measurement, the air temperature was 20.4 ± 0.3°C and D was 0.009 ± 0.0008.

Measurement of bark traits

After measuring g_{bark}, bark traits were measured on each stem segment. Stem diameter and bark thickness were measured with calipers on both the distal and basal ends of each segment. Bark included all tissue radially outward of the vascular cambium. The bark thickness ratio was calculated as two times the bark thickness divided by stem diameter (Hoffmann et al. 2012).~~After measuring g_{bark}, bark traits were measured on each stem segment. Stem diameter and bark thickness were measured with calipers on both the distal and basal ends of each segment. The bark thickness ratio was calculated as two times the bark thickness divided by~~

~~stem diameter.~~

For the g_{bark} survey samples, high-resolution images of the stem segments were taken using a Canon CanoScan LiDE 400 flatbed scanner. Within a rectangular area on the surface of each sample, the total number of lenticels was counted, and the area of each lenticel (a measurement of lenticel size) was measured using ImageJ (Schneider et al. 2012). For each segment, lenticel density was calculated as the number of lenticels divided by the area of the stem surface measured and lenticel size was taken as the average area of the lenticels. ~~Lenticel density for each segment was calculated as the number of lenticels divided by the area of the stem surface measured. Lenticel area was averaged within each segment.~~

For the flooding experiment, lenticels were tallied on the exposed area of the stem segment not covered by the parafilm, and lenticel density was calculated by dividing the total number of lenticels by the exposed surface area. Lenticel width (a proxy for lenticel size, perpendicular to the longest dimension of a lenticel) was measured on 10 randomly selected lenticels on each segment with calipers under a 10× lens and averaged for each segment. All lenticel measurements were performed on primary lenticels, whereas secondary lenticels, which are often obscured by successive periderm layers (Rosner and Morris 2022), were excluded due to the difficulty of observing them. ~~For the flooding experiment, lenticels were tallied on the entire exposed area of the stem segment, and lenticel density was calculated by dividing the total number of lenticels by the exposed surface area. Lenticel vertical width (a measurement of lenticel size) was measured on 10 randomly selected lenticels on each segment with calipers~~

~~under a 10× lens and averaged for each segment.~~

Data analysis

All analyses were performed in R version 4.2.2 (R Core Team 2022). We transformed the response variable when needed to meet the underlying model assumptions. We set the significance level (alpha) to 0.05.

For the g_{bark} survey, a two-way ANOVA was performed to test the effects of species and segment position on g_{bark}. The response variable was square root transformed g_{bark} and the explanatory variables were species, segment position, and their interaction. Two ~~P~~planned contrasts were used to compare the means of g_{bark}; one compared among species and one compared among segment positions~~ among species and segment positions~~. We tested for relationships between species-level bark traits and g_{bark} using ordinary least squares regression. We also used ordinary least squares regression to test for relationships within species between g_{bark} and traits. For eight species with enough data on lenticel density and lenticel size (i.e., at least five out of the nine stem segments measured had at least one lenticel), the response variable was g_{bark}, and the explanatory variables were lenticel size, lenticel density, the interaction of lenticel size and lenticel density, and bark thickness ratio. For the other seven species without enough data on lenticel traits, the response variable was g_{bark}, and the explanatory variable was bark thickness ratio only.

For the bending experiment, we performed a three-way ANOVA with an error term to account~~was performed that accounted~~ for the split-plot design, where species served as the whole plot factor, and bending treatments were the split-plot factor, which was nested in by the segment positions. The response variable was the ~~with~~ natural log transformed g_{bark}, with ~~as the response variable, and~~ bending treatment, species, segment position, and their interaction, included as the fixed effects.~~explanatory variables.~~ Planned contrasts were used to compare g_{bark} between the two levels of bending treatment, among species, and across segment positions~~position~~.

For the flooding experiment, we performed a three-way ANOVA to test the effects of flooding treatment, species, and segment positions on g_{bark}. The response variable was natural log transformed g_{bark}, and the explanatory variables were flooding treatment, species, segment position, and their interaction. We used planned contrasts to compare g_{bark} among post-flood treatments, species, and segment positions. The relationships between g_{bark} and lenticel density, lenticel size, and bark thickness ratio for each species were tested by ordinary least squares regression. The response variable was natural log transformed g_{bark} for *L. styraciflua* and *N. aquatica* separately, and the explanatory variables were lenticel density, lenticel size, and bark thickness ratio. Additionally, for each species, we performed ANOVA to test the effect of flooding treatment on lenticel density and lenticel size. The response variables were lenticel density or lenticel size of *L. styraciflua* and *N. aquatica* separately, and the explanatory variable was flooding treatments. If the interactions are not significant for the tests, we reported the p-values of the final models without interaction.

Results

g_{bark} survey

Among 135 stem segments measured in the g_{bark} survey, the mean g_{bark} was 3.84 ± 1.43 mmol m^{-2} s^{-1}, with a range of 0.87 to 9.20 mmol m^{-2} s^{-1}. g_{bark} varied among species and segment positions (Fig. 3; $F_{14,90} = 12.80$, p~~P~~ < 0.001; and $F_{2,90} = 5.54$, p~~P~~ < 0.01; respectively), and not their interaction ($F_{28,90} = 0.61$, p~~P~~ $= 0.93$). Among species, g_{bark} ranged from 2.22 ± 0.93 mmol m^{-2} s^{-1} in *Fagus grandifolia* to 6.31 ± 1.69 mmol m^{-2} s^{-1} in *Acer negundo*. Segment positions ranged in g_{bark} from 3.48 ± 1.23 mmol m^{-2} s^{-1} in the distal position (closest from the apex) to 4.11 ± 1.53 mmol m^{-2} s^{-1} in the proximal position (furthest from the apex) (Fig. 4). Among species, g_{bark} did not scale with lenticel density, lenticel size, or bark thickness ratio (Fig. S1).

The intraspecific relationships between g_{bark} and bark traits differed among species (Table S1). Only two variables were significantly related to g_{bark}; bark thickness ratio for *Acer negundo* ($P = 0.04$) and lenticel density for *Betula nigra* ($P = 0.02$). Both were negatively related to g_{bark} For seven out of the eight species on which we encountered lenticels, lenticel density trended negatively with g_{bark}. No consistent trends were found for lenticel size or bark thickness ratio.

Bending experiment

In the bending experiment, across all species and segment positions, g_{bark} averaged 5.55 ± 2.19 mmol m^{-2} s^{-1}, with a range of 2.66 to 13.62 mmol m^{-2} s^{-1}. The ANOVA showed that g_{bark} was significantly affected by bending treatment and species, but not segment position or any interactions (Fig. 5; $F_{1,59}$ = 4.44, p~~P~~ = 0.03; $F_{3,59}$ = 10.57, p~~P~~ < 0.001; $F_{2,59}$ = 0.67, p~~P~~ = 0.52; and $F_{6,59}$ = 0.16, p~~P~~ = 0.99; respectively). On average, there was an increase of 1.13 mmol m^{-2} s^{-1} g_{bark} from~~between~~ the~~ bent and~~ control segments to the bent segments (Fig. 5A). g_{bark} was similar across species except for *P. occidentalis*, where it was significantly lower (Fig. 5B).

Flooding experiment

Among all segments of both species in the flooding experiment, g_{bark} averaged 8.10 ± 7.44 mmol m^{-2} s^{-1} with a range of 1.54 to 51.21 mmol m^{-2} s^{-1} (Fig. 6). g_{bark} differed between species (Fig. 6B; $F_{1,62}$ = 116.34, p~~P~~ < 0.001), with values of 4.37 ± 2.96 mmol m^{-2} s^{-1} in *N. aquatica* and 12.02 ± 8.68 mmol m^{-2} s^{-1} in *L. styraciflua*. g_{bark} varied among post-flood durations (Fig. 6A; $F_{3,62}$ = 4.43, p~~P~~ < 0.01). It ranged from 6.32 ± 4.45 mmol m^{-2} s^{-1} in the 69-day post-flood group to 10.70 ± 11.30 mmol m^{-2} s^{-1} in the control group. g_{bark} also varied between segment positions (Fig. 6C; $F_{1,62}$ = 62.05, p~~P~~ < 0.001), with proximal stem segments, on average, over twice as high as distal stem segments (11.10 ± 9.09 versus 5.10 ± 3.34 mmol m^{-2} s^{-1}). No interactions were significant.

The relationships between g_{bark} and bark traits differed between species (Table 3). In *L. styraciflua*, g_{bark} decreased as lenticel density increased (t_{36}~~t~~ = -3.92, p~~67, P~~ < 0.001). In *N. aquatica*, g_{bark} decreased as the bark thickness ratio increased (t_{38} = -4.76, p~~t = -3.98, P~~ < 0.001). Whereas g_{bark} did not scale with other traits, the decreasing trend in g_{bark} when the bark thickness ratio increased was consistent for both species (~~Table 3~~Table 3).

During the flooding experiment, the lenticel size of *L. styraciflua* significantly differed among flooding treatments ($F_{3,34}$ = 5.83, p~~82, P~~ < 0.01). The lenticel size of *L. styraciflua* was 0.44 ± 0.12 mm in the control group, 0.57 ± 0.12 mm in the 0 day post-flood duration group, 0.64 ± 0.15 mm in the 42 days post-flood duration group, and 0.68 ± 0.20 mm in the 69 days post-flood duration group. The lenticel density of *L. styraciflua* and lenticel size and lenticel density of *N. aquatica* did not differ among flooding treatments.

Discussion

g_{bark} *variation among species*

Tree species differed in g_{bark} within our survey in a temperate deciduous forest in the southeastern USA and within both of our experiments. In our survey, the mean and range of g_{bark} (Fig. 1) were similar to those of previous studies in various biomes (~~Table 1~~Table 1). Consistently, g_{bark} variation is high among species within biomes and low among biomes, indicating that biome is not informative in predicting g_{bark}. Nonetheless, g_{bark} contains phylogenetic signals (Ávila-Lovera and Winter 2024) and is an informative species-level trait (Wolfe 2020, Loram-Lourenço et al. 2022). However, like other functional traits, g_{bark} varies within individuals and in response to environmental conditions. Among co-occurring species within biomes, the variation in g_{bark} along with other traits indicates different ecological strategies (niches) to avoid interspecies competition and maintain diversity (Chase and Leibold 2003, Kraft et al. 2008).

g_{bark} *variation along stems*

Our survey and flooding experiment both showed that stem segments further from the apex had higher g_{bark} (Figs. 4, 6C). In contrast, the bending experiment found no differences in g_{bark} among segment positions (Fig 5C). However, there was a trend for lower g_{bark} in distal segments in the bending experiment, so the lack of effect is probably due to a smaller sample size. In contrast to our results, Wolfe (2020) found a weak and marginally significant trend for g_{bark} to decline from the distal to proximal positions of tropical tree saplings, from 3.85 mmol m^{-2} s^{-1} at 4 cm from the apex to 2.94 mmol m^{-2} s^{-1} at 250 cm from the apex. Whereas we measured terminal branches of large trees in the survey and bending experiment and small saplings in the flooding experiment, Wolfe (2020) measured across the entire stem length of large saplings. The factors that underlie these divergent results are unclear; however, they may be related to ontogeny and sampling design.

Associations between bark traits and g_{bark}

We found no clear relationship between g_{bark} and lenticel density. Among tree species in our survey, g_{bark} was unrelated to lenticel density (Fig. S1), whereas within most species there was an insignificant trend for a negative relationship between g_{bark} and lenticel density (Table S1). In the flooding experiment, the relationships between g_{bark} and bark traits were not consistent, with only lenticel density of *L. styraciflua* and bark thickness ratio of *N. aquatica* being significantly related to g_{bark}. Lenticel size was inversely related or showed a negative trend with lenticel density in most species (Table S2). Lenticel size did not affect g_{bark} in any of the species. However, Rosner and Morris (2022) suggested that lenticel size was positively related to g_{bark} in *Picea abies*. Overall, our~~Overall, these~~ results contrast those of Loram-Lourenço et al.

(2022), who found that g_{bark} was negatively related to relative outer bark thickness and positively related to lenticel density among terminal branches of 10 tree species in a Brazilian savanna.

Even though substantial evidence indicates lenticels are more permeable to water vapor and oxygen than phellem areas without lenticels (Schönherr and Ziegler 1980, Groh et al. 2002) and lenticels facilitate tree survival during long-term flooding (Larson et al. 1991), plants regulate lenticel properties seasonally (Lendzian 2006), which implies the mechanisms behind lenticels controlling bark gas exchange are complex and not sufficiently explored. The considerable variability of lenticel structure and abundance and the probability of waxes covering the inner lenticel surface may weaken the associations between lenticel traits and g_{bark} (Rosner and Morris 2022).~~The considerable variability of lenticel structure and abundance and the probability of waxes covering the inner lenticel surface may weaken the associations between lenticel traits and~~ g_{bark} ~~(Rosner and Morris 2022).~~ We measured g_{bark} outside of the growing season, when lenticels may have been partly occluded by an intact, densely packed, and suberized cell layer (Rosner and Morris 2022),~~;~~ which would diminish the relationship between g_{bark} and lenticel density and size. A similar structure was found on potato tuber lenticels to reduce the conductance to water vapor (Hayward 1974).~~.~~ However, the seasonality of g_{bark} and its relationship with lenticel structure is not clear and requires further research. Moreover, as rhytidome forms, cell division within primary lenticels often slows, causing the lenticels to become less distinct (Crang et al. 2018), which would further reduce their relationship with g_{bark}. Future studies that control for bark structure rather than distance from the apex would have more ability to detect relationships between g_{bark} and lenticel traits.

In the flooding experiment, g_{bark} decreased with higher bark thickness ratio (Table 3). This result is similar to those of Wolfe (2020), Loram-Lourenço et al. (2022), and Ávila-Lovera and Winter (2024). Commonly, bark permeability is thought to decrease with bark thickness (Paine et al. 2010). Tree species that grow in drier and hotter environments tend to have thicker bark (Rosell 2016), which may reflect adaptation for reducing water loss. However, our g_{bark} survey did not find consistent relationships between the bark thickness ratio and g_{bark}, within or among species (Fig. S1, Table S1), which contrasted with the previous studies (Wolfe 2020, Loram-Lourenço et al. 2022, Ávila-Lovera and Winter 2024), probably because the average bark thickness ratio of trees from the g_{bark} survey was consistently low, 0.09 ± 0.03 with a variance of 0.0008. The limited variability of bark thickness ratio of the g_{bark} survey made it difficult to detect significant relationships. Also, Loram-Lourenço et al. (2022) measured the outer bark thickness, which is the barrier to water vapor and is expected to relate more closely to g_{bark} compared to the total bark thickness in the current survey. Ávila-Lovera and Winter (2024) found that g_{bark} decreased with bark thickness ratio among tropical tree species, however, they measured g_{bark} in stem segments located 10 cm from the apex. This placement presumably included areas of primary growth, as indicated by the presence of stomata described by the authors, as opposed to the secondary growth segments we collected, which may have led to the divergent results.

Effects of stem bending on g_{bark}

Directly after stems were bent, they had significantly higher g_{bark} than unbent controls (Fig. 5A). Although no macroscopic bark damage was observed, it is possible that bending caused undetected damage to the outer bark that was obscure or microscopic. For example, compression and expansion of suberin and wax layers in periderm cells (including those in the lenticels) or abrasion between successive layers in rhytidome could have increased g_{bark}. Similar to our results, when leaves were subjected to wind, cuticle damage caused higher minimum vapor conductance in conifer trees (van Gardingen et al. 1991). It is unclear to what extent our bending experiment simulated stem bending caused by wind; however, considering that none of the stems broke during the experiment and that stem breakage is a common response to cyclones (Gleason et al. 2008), our experiment likely captured realistic bending stress that occurs during windstorms. Wind damage can affect forest vulnerability to subsequent droughts in multiple ways; for example, when hurricanes create canopy gaps, environmental conditions favor fast-growing tree species that are vulnerable to drought (Smith-Martin et al. 2022). Our results suggest that increased g_{bark} is an additional mechanism by which windstorms can make forests vulnerable to droughts. All else equal, residual water loss from trees will increase in proportion to g_{bark} and the drought timespan to reach critical thresholds of stem water potential will decrease. Distinguishing the wind conditions that affect g_{bark} and how they vary among trees will require further research.

Effects of flooding on g_{bark} and lenticels

Contrary to our expectations, we found that 38 days of flooding did not cause increased g_{bark} (Fig 6A). Instead, g_{bark} decreased during the recovery process (Fig. 6A). Based on extensive observations of flooding causing saplings to form hypertrophied lenticels near the waterline on their stems (Esau 1965, Schaffer 1998, Shimamura et al. 2010)~~(Esau 1965, Schaffer 1998, Shimamura et al. 2010)~~, and evidence that hypertrophied lenticels are highly permeable (Rennenberg et al. 1997)~~(Rennenberg et al. 1997)~~, we expected proximal stem segments to have particularly high g_{bark} in response to flooding. However, the lack of this response may reflect that the saplings were well adapted to the low soil oxygen availability in our flood treatment, which would preclude a stress response such as increased g_{bark}. *L. styraciflua* and *N. aquatica* are described as moderately flood-tolerant (survives waterlogging or saturated soils for 30 consecutive days during the growing season)~~-~~ and extremely flood-tolerant (survives deep, prolonged waterlogging for more than one year),~~,~~ respectively (Niinemets and Valladares 2006). Constitutively high g_{bark} near the stem base (i.e., proximal position) may be an adaptation for flood tolerance. Since hypertrophied lenticels develop over several days or weeks, constitutively high g_{bark} would help to reduce root oxidative stress at the onset of soil waterlogging. This is supported by the result that g_{bark} was higher in proximal segments than distal segments across all treatments (Fig. 6C). This result contrasts with those of Wolfe (2020), who found that g_{bark} varied only marginally from distal to the proximal positions of saplings in seasonally dry tropical forests that are not prone to flooding. The proximal position of these two flood-tolerant species also had relatively high g_{bark} compared to the saplings measured by Wolfe (2020), which is the only other report of g_{bark} in saplings that we are aware of. However, constitutively high g_{bark} would be disadvantageous during drought conditions, which may contribute to the tradeoff between flood tolerance and drought tolerance among tree species (Niinemets and Valladares 2006).

The effect of flooding on lenticels varied between the two tree species. In *L. styraciflua*, lenticel size increased after the 38-day flooding period and continuously increased with a longer post-flood duration, with no corresponding increase in g_{bark}.~~.~~ This is consistent with results from Angeles et al. (1986), who observed the formation of hypertrophied lenticels in *Ulmus americana* seedlings as soon as five days after~~of~~ flooding. The morphological changes included more active phellogen and elongated cork, phellem, and cells, enhancing the gas exchange rate of the bark (Angeles et al. 1986). No changes were found in the lenticel size in *N. aquatica*, which likely possesses adaptations that enable it to grow in flooded environments without morphological changes like lenticel hypertrophy.

After a period of 69 days post-flood, g_{bark} declined below that of the control plants (Fig. 6A), contrasting with our expectations. The processes that drove this g_{bark} decline are unclear. One possibility is that the flood treatment induced early dormancy in the saplings. In many tree species, flooding causes early leaf senescence and physiological dormancy (Pereira and Kozlowski 1977, Vu and Yelenosky 1991). Winter dormancy is associated with the formation of a lenticel closing layer composed of densely packed suberized cells~~occlusion~~ (Langenfeld-Heyser 1997), which would reduce g_{bark}. Therefore, the saplings that were exposed to flooding in the early summer months could have entered a dormant state by the time they were measured in the early fall, while the control saplings were still active with unoccluded lenticels leading to the observed higher g_{bark}. Since lower g_{bark} is associated with higher drought performance (Wolfe 2020, Loram-Lourenço et al. 2022), the reduced g_{bark} in the post-flood treatment would be expected to enhance drought performance compared to the control trees that were not exposed to flood conditions. However, it is unclear how long the trees would maintain a post-flood g_{bark} reduction or whether other effects of flooding would diminish drought performance.

Conclusions

We found that g_{bark} varied widely among tree species in a temperate forest, consistent with studies in a wide range of biomes. The lack of consistent relationships between g_{bark} and bark traits such as lenticel density, lenticel size, and bark thickness point to more complex drivers of g_{bark} variation such as the structure and composition of periderms and lenticels and their dynamics. Stem bending and flooding both affected g_{bark}. This suggests that g_{bark} dynamics can mediate the interactive effects of extreme weather events on trees and forests. Extreme weather events are likely to become more common this century (Beniston et al. 2007, Konisky et al. 2015), so resolving the factors that lead to g_{bark} variation, its dynamics, and its impacts on tree performance should be a research priority.

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Authors' contributions

B.T.W., G.R., and Y.Z designed and performed the research; Y.Z., B.T.W., and V.D. analyzed the data. Y.Z. wrote the paper and B.T.W. revised it. All authors reviewed and approved the final manuscript.

Data availability statement

Data are available from the corresponding author upon request.

Conflict of interest

The authors declare that they have no conflict of interest.

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Figure legends

Figure 1 Schematic figure for bending experiment. Two ~1 m length terminal branches were collected from three trees of four species. One branch from each tree was subjected to an unbent control treatment (A), referred to as "control". The frame was rolled along the branch to account for potential abrasion. The other branch was subjected to a bending treatment (B), referred to as "bent". We bent the branch 10 times around the frame and allowed it to return to its straight position. The treatments were applied for 10 times. The branch was rotated between each treatment to ensure evenly distributed pressure.~~**Figure 1** Schematic figure for bending experiment. Panel (A) indicates the control group and panel (B) indicates the treatment (bent) group.~~

Figure 2 Schematic figure for flooding experiment. Panel (A) indicates the control treatment~~group~~ and panel (B) indicates the flooding treatments~~treatment groups~~ with consistent flood durations and variable post-flood durations. Five saplings of each species were placed in each round of flooding and the control. In total 40 saplings were applied with one *L. styraciflua* sapling was lost in the Round 2 flooding treatment; thus 39 saplings were harvested. The saplings experienced different lengths of time post-flood and were measured for g_{bark}.

Figure 3 Boxplot of bark vapor conductance (g_{bark}) among tree species surveyed in a temperate forest. Each box extends to 25% and 75% quartiles of values and is bisected by the median. Whiskers extend to the most extreme values within 1.5 times the interquartile range.

Circles outside of the whiskers represent outliers. Differing letters above each boxplot represent statistically different square root transformed g_{bark} values between combinations, based on planned contrasts.

Figure 4 Boxplot of bark vapor conductance (g_{bark}) among stem segment positions surveyed in a temperate forest. Boxes, whiskers, and differing letters are drawn as in Fig. 3. Blue lines represent pairs that had higher g_{bark} in the proximal segments than in the distal segments, and red lines, vice versa. Differing letters above each boxplot represent statistically different square root transformed g_{bark} values between combinations, based on planned contrasts.

Figure 5 Boxplots of bark vapor conductance (g_{bark}) in the bending experiment grouped by (A) treatment, (B) species, and (C) segment position. Boxes and whiskers are drawn as in Fig. 3. In panel (A) lines connect g_{bark} values of segment pairs (i.e., same tree and segment position). Blue lines represent pairs that had higher g_{bark} in the bent treatment than in the control treatment and red lines, vice versa. Differing letters above each boxplot represent statistically different natural log transformed g_{bark} values between combinations, based on planned contrasts. The letters "NS" in panel (C) indicate natural log transformed g_{bark} did not differ between segment positions in this experiment.

Figure 6 Boxplots of bark vapor conductance (g_{bark}) in the flood experiment grouped by (A) treatment, (B) species, and (C) segment position. Boxes and whiskers are drawn as in Fig. 3.

Differing letters above each boxplot represent statistically different natural log transformed g_{bark} values between combinations, based on planned contrasts.

Table 1 Comparison of g_{bark} among biomes. *NA* indicates that original data were not present or not available. g_{bark} range and mean (mmol m^{-2} s^{-1}) present the range and mean of species means, respectively.

Source	Biome	Location	Number of species	g_{bark} range (mmol m^{-2} s^{-1})	g_{bark} mean (mmol m^{-2} s^{-1})
g_{bark} survery in present study	Temperate forest	Louisiana, USA	15	2.22-12.02	3.84
Ávila-Lovera and Winter (2024)	Tropical forest	Panama	94	0.94-24.60	8.77
Ávila-Lovera et al. (2021)	Tropical forest	Miranda, Venezuela	1[a]	20.5-34.6	27.68
Loram-Lourenço et al. (2022)	Tropical savanna	Goiás, Brazil	10	2.97-11.0	7.22
Wolfe (2020)	Tropical forest	Panama	8	1.68-8.57	5.84
Ávila-Lovera et al. (2017)[b]	Desert-chaparral	California, USA	11	1.11-20.49	6.69
Wittmann and Pfanz (2008)	Temperate forest	Duisburg, Germany	5	5.01-27.3	12. 92[c]
Cernusak and Marshall (2000)	Mixed-conifer temperate forest	Idaho, USA	1	*NA*	1.03

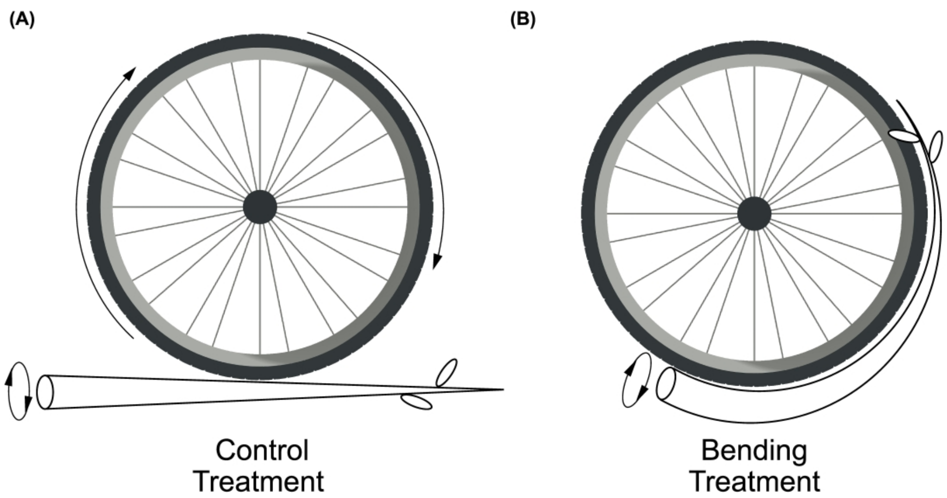
[a]Three cultivars of cacao were measured
[b]Only including saplings in forests.Mean of species means
[c]Mean of wet and dry seasons

Table 2 Species sampled in g_{bark} survey. Leaf phenology descriptions are from local observation.

Common name	Scientific name	Family	Leaf phenology
Boxelder	*Acer negundo*	Sapindaceae	Deciduous
River birch	*Betula nigra*	Betulaceae	Deciduous
Pecan	*Carya illinoiensis*	Juglandaceae	Deciduous
Sugarberry	*Celtis laevigata*	Cannabaceae	Deciduous
American beech	*Fagus grandifolia*	Fagaceae	Deciduous
Southern magnolia	*Magnolia grandiflora*	Magnoliaceae	Evergreen
Water tupelo	*Nyssa aquatica*	Nyssaceae	Deciduous
Slash pine	*Pinus elliotti*	Pinaceae	Evergreen
Loblolly pine	*Pinus taeda*	Pinaceae	Evergreen
American sycamore	*Platanus occidentalis*	Platanaceae	Deciduous
Southern red oak	*Quercus falcata*	Fagaceae	Deciduous
Cherrybark oak	*Quercus pagoda*	Fagaceae	Deciduous
Bald cypress	*Taxodium distichum*	Cupressaceae	Deciduous
Winged elm	*Ulmus alata*	Ulmaceae	Deciduous
American elm	*Ulmus americana*	Ulmaceae	Deciduous

Table 3 Slope coefficients from least squares regressions between g_{bark} and bark traits. The coefficients in bold and marked with *** had $p < 0.001$, otherwise $p > 0.05$.

Species	Lenticel density	Lenticel size	Bark thickness ratio
Liquidambar styraciflua	**-0.05*****	-0.39	-6.02
Nyssa aquatica	0.004	0.03	**-9.60*****



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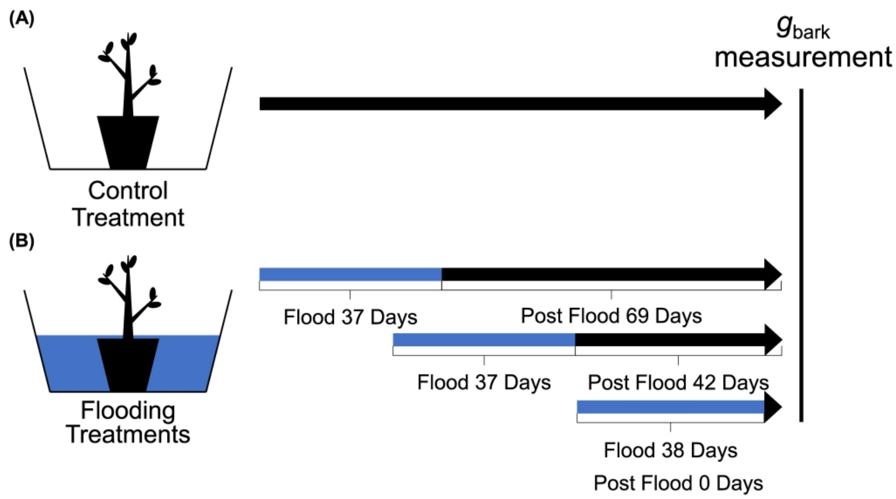

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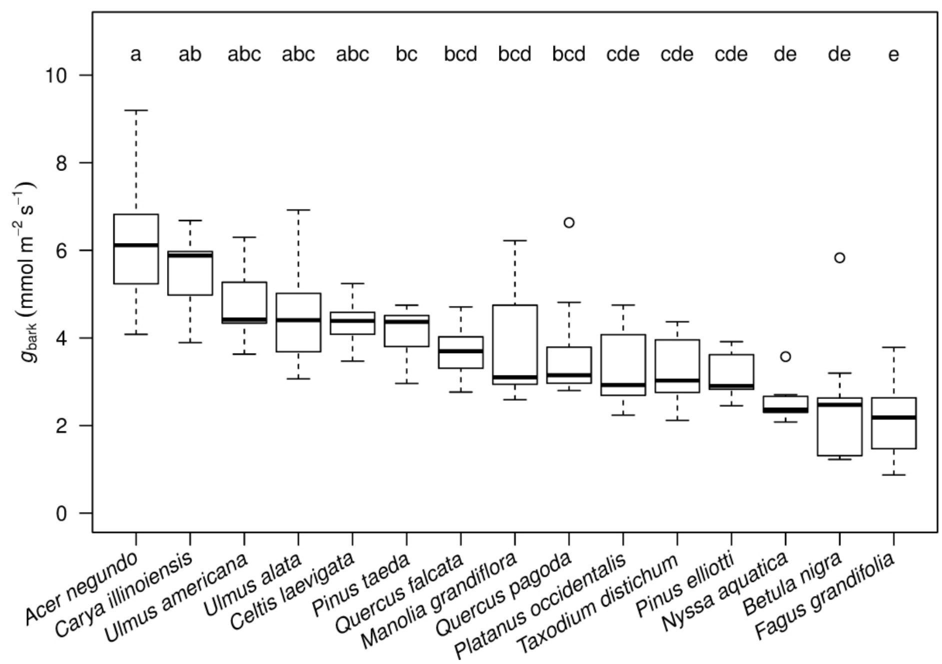

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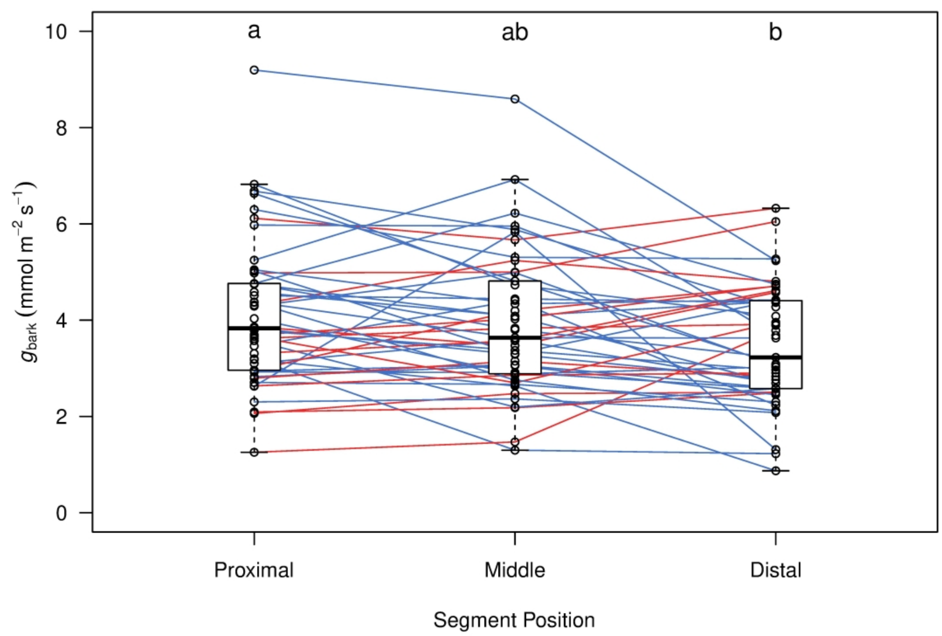

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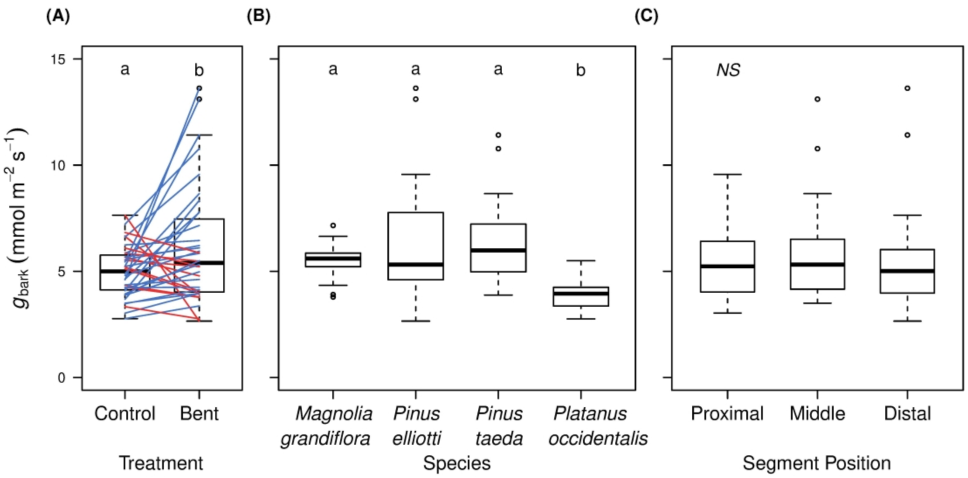

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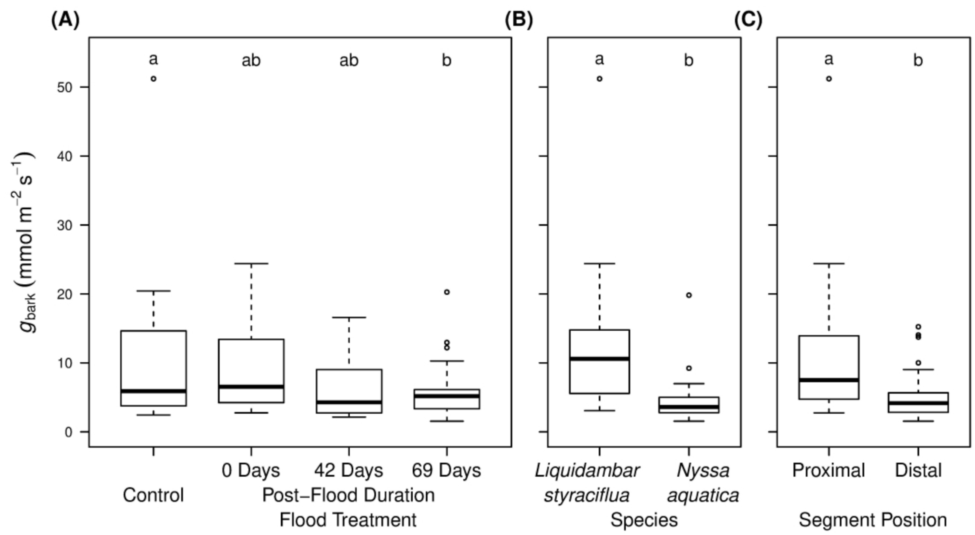

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